

Assignment Report

Initial Steps

Turtle Games are an international business that manufactures and sells a variety of entertainment products like books, board games and toys. Turtle Games are looking to find out how their customers accumulate loyalty points, how their customers can be best segmented and how their textual data can be utilised all to improve marketing campaigns by making them more targeted, and by making the business more efficient.

The approach I took to import, clean and then analyse my data was first off to get a sense of the data set I was working with using pandas:

```
print(reviews.head())
print(reviews.info())
reviews.describe()
reviews.isnull().values.any()
```

This is to see the first 5 rows of the data set, then to gather metadata and initial descriptive statistics on the numeric columns. The last line is to check whether there are any nulls.

A copy of the dataset was created to preserve the raw data, and irrelevant columns were dropped (for instance, *language* and *platform* were excluded since all values were constant). Columns were also renamed for consistency using:

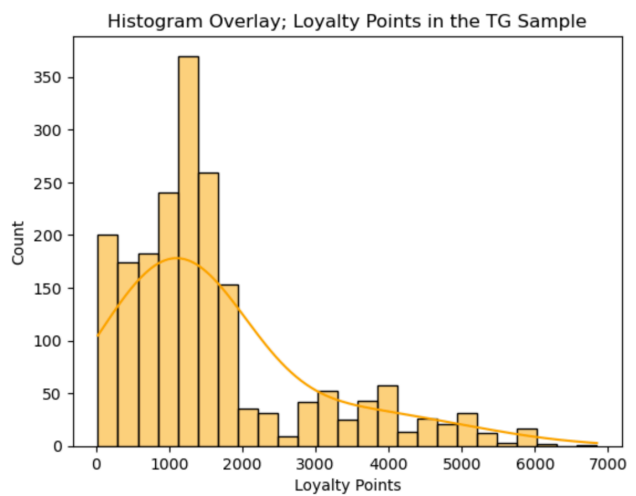
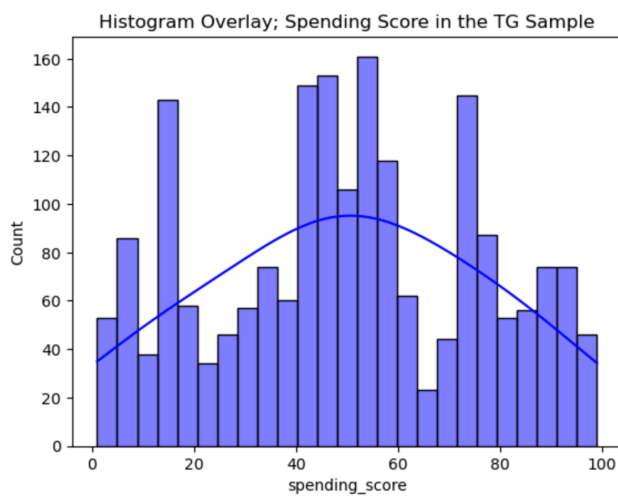
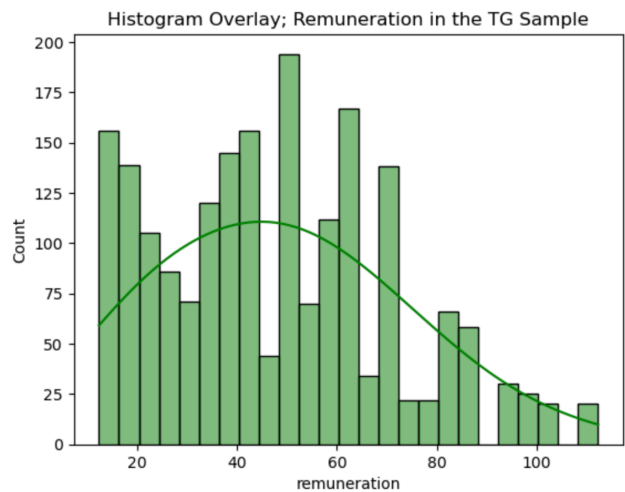
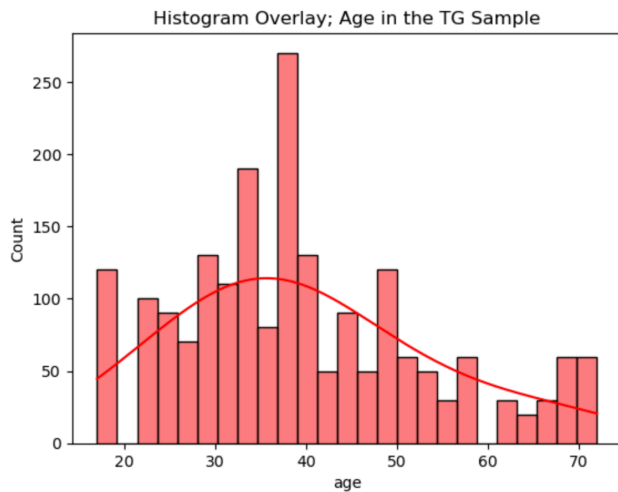
```
turtle_reviews_clean.rename(
columns = {'remuneration (k£)': 'remuneration', 'spending_score
(1-100)': 'spending_score'}, inplace = True)
```

The above snippet was used to rename columns, with inplace set to true so that the changes to the dataframe were permanent in effect. Then re view the data frame, save the clean version as a CSV to the working directory, re-import and repeat the initial steps. Categorical variables such as *gender*, *product*, and *education* were transformed into dummy variables with `pd.get_dummies()` To ensure compatibility with machine learning models.

Evaluating the data

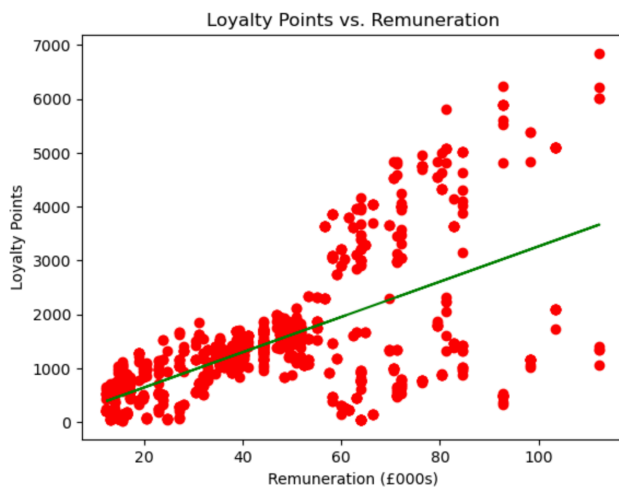
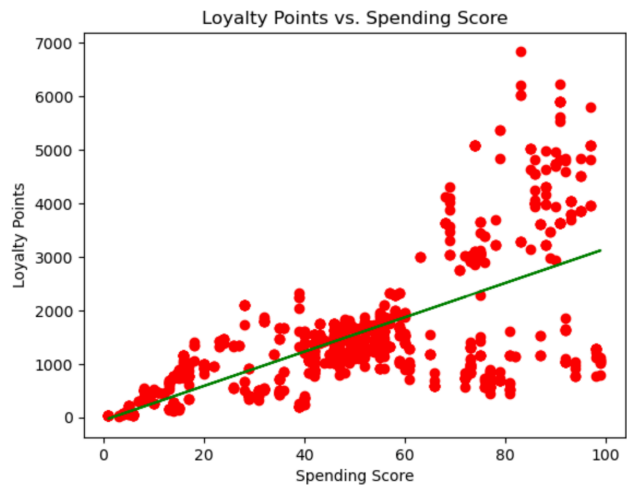
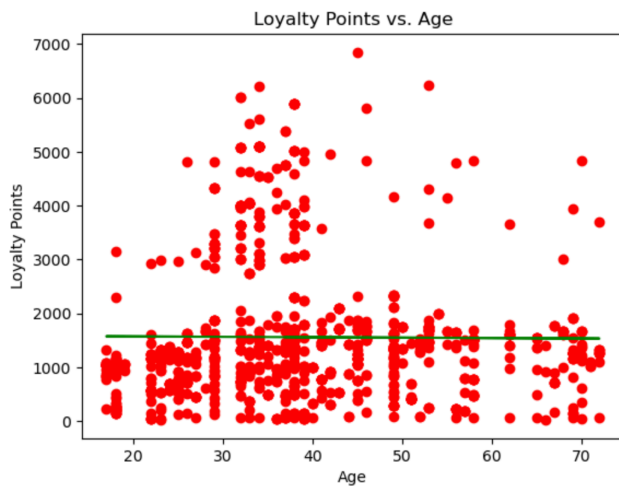
Following the initial cleaning of the data, the dataset was then evaluated for normality, skewness, and kurtosis to check to see how close the distribution was to a normal distribution. This was done so that there wouldn't be any undue bias in the could affect results and so compromise the overall analysis. In the graphs below I visualised skewness and kurtosis for a more intuitive look for each of the numeric variables. Loyalty points had a high kurtosis and strong skew, so a **logarithmic transformation** was applied to stabilise variance and normalise the distribution:

```
reviews_clean["log_loyalty_points"] =
np.log1p(reviews_clean["loyalty_points"])
```



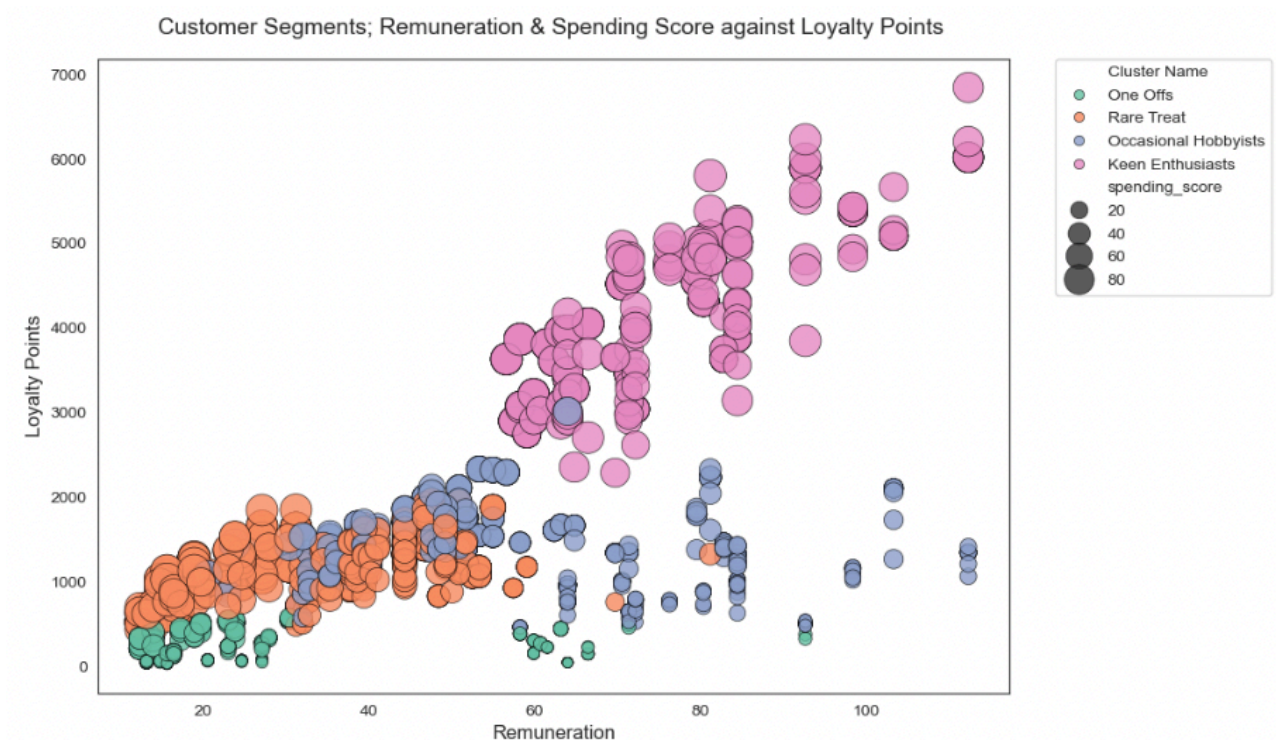
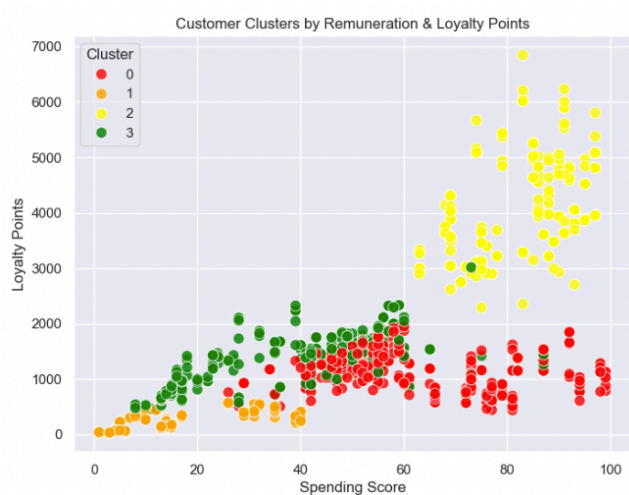
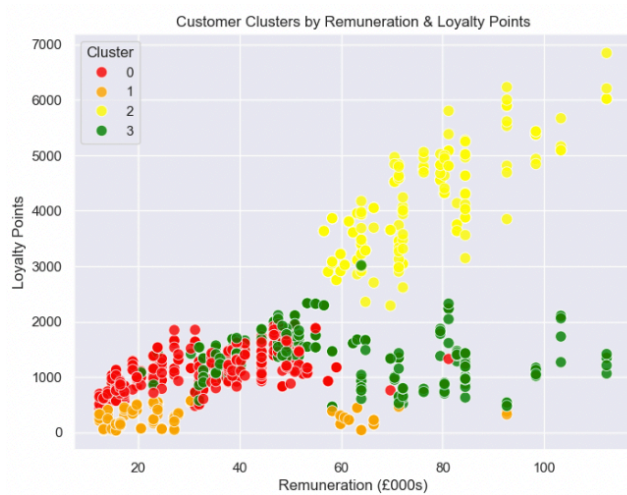
Linear Regression

With the data properly cleaned and adjusted the next step was to perform linear regression on the numeric columns in the data set to see how each of them influence loyalty points, as in the visuals below. For multiple linear regression the value returned for the R squared was 0.83, so with remuneration and spending score, so the confidence using these predictors is high. The results were further verified using OLS regression tests. I then used a Decision Tree Regressor and Random Forest Regressor to further validate my results.



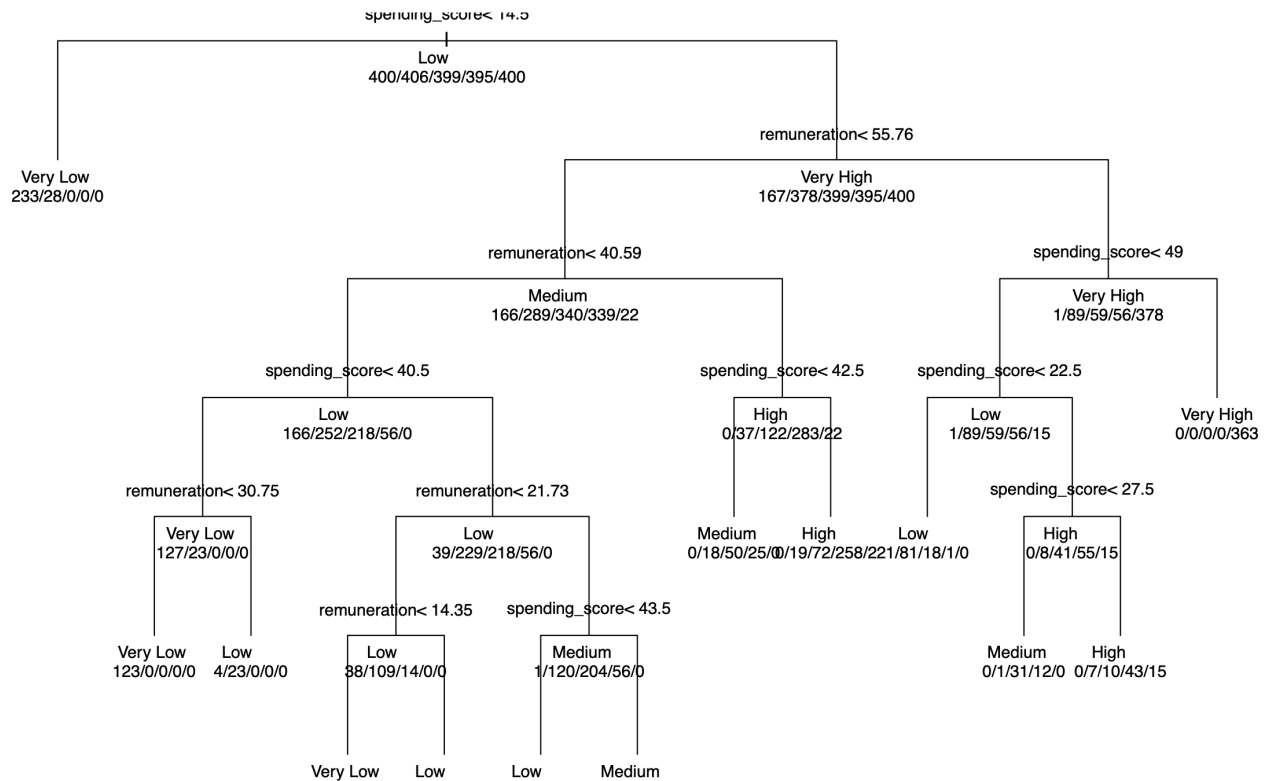
Clustering/Segmentation

With regards to customer segmentation I used k-means clustering and scatter plots to visualise my findings. This was done to depict distinct customer groups, with each cluster representing a specific customer type with a creative name for easier memorisation. This way, the insights drawn could be utilised directly in marketing & strategy recommendations. Overall, the combination of statistical and visual analysis provided a clear, understanding of Turtle Games's customer base and offered strategic guidance for loyalty and engagement initiatives. The model has 3 dimensions; Remuneration on the x axis and spending score as bubble size. This serves to make the visual more immediately and intuitively perceivable; 4 clusters were chosen given the silhouette and elbow method.



Also, classification analysis in R supported the findings from Python; reinforcing our discovery that remuneration and spending score are the strongest predictors of loyalty points (below).

Classification Tree for Loyalty Points



Textual Analysis

For the textual analysis I used the VADER model to analyse the sentiment of the reviews in the data. The new data was then added to the current dataframe and analysed via histogram and trendline; results were widely very positive with some more of the reviews appearing in the neutral side of the scale. Then, to create the wordcloud, first all of the text was made lowercase, stopwords and punctuation removed, then created using Seaborn and wordcloud. The final step for textual analysis was to find the most frequent words used by those who left reviews, producing both a list count and a more interpretable bar graph.

